THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012 Fundamental Concepts of Mathematics

Solution to **Assignment 2**

1. (a) $(\sim P) \vee (\sim Q)$

P	Q	$\sim P$	$\sim Q$	$(\sim P) \vee (\sim Q)$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

(b) $\sim (P \vee Q)$

P	Q	$P \vee Q$	$\sim (P \vee Q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

(c) $(\sim P) \wedge ((\sim Q) \vee (\sim R))$

P	Q	R	$\sim P$	$\sim Q$	$\sim R$	$(\sim\leftrightarrow Q) \vee (\sim R)$	$(\sim P) \wedge ((\sim Q) \vee (\sim R))$
T	T	T	F	F	F	F	F
T	T	F	F	F	T	T	F
T	F	T	F	T	F	T	F
T	F	F	F	T	T	T	F
F	T	T	T	F	F	F	F
F	T	F	T	F	T	T	T
F	F	T	T	T	F	T	T
F	F	F	T	T	T	T	T

(d) $P \to (Q \to R)$

P	Q	R	$Q \to R$	$P \to (Q \to R)$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

(e) $(P \leftrightarrow Q) \vee (P \rightarrow R)$

P	Q	R	$P \leftrightarrow Q$	$P \rightarrow R$	$(P \leftrightarrow Q) \vee (P \rightarrow R)$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	T	T
F	T	F	F	T	T
F	F	T	T	T	T
F	F	F	T	T	T

(f) $(P \leftrightarrow Q) \wedge (R \wedge P)$

P	Q	R	$P \leftrightarrow Q$	$R \wedge P$	$(P \leftrightarrow Q) \wedge (R \wedge P)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	T	F
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	F	F
F	F	T	T	F	F
F	F	F	T	F	F

2. (a) We only need to show that $((P \rightarrow Q) \wedge P) \rightarrow Q \Longleftrightarrow \mathbf{T}$:

$((P \rightarrow Q) \wedge P) \rightarrow Q \Leftrightarrow \sim ((P \rightarrow Q) \wedge P) \vee Q \Leftrightarrow \sim (((\sim P) \vee Q) \wedge P) \vee Q \Leftrightarrow \sim (\mathbf{F} \vee (Q \wedge P)) \vee Q$
$\Leftrightarrow \sim (Q \wedge P) \vee Q \Leftrightarrow (\sim Q \vee \sim P) \vee Q \Leftrightarrow \mathbf{T} \vee (\sim P \vee Q) \Leftrightarrow \mathbf{T}$.

Alternatively, we may consider the following truth table:

P	Q	$(P \rightarrow Q)$	$(P \rightarrow Q) \wedge P$	$((P \rightarrow Q) \wedge P) \rightarrow Q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

So the given statement is always true regardless of the truth values of P and Q.

(b) We only need to show that $((P \vee Q) \wedge (\sim P)) \rightarrow Q \Longleftrightarrow \mathbf{T}$:

$((P \vee Q) \wedge (\sim P)) \rightarrow Q \Leftrightarrow \mathbf{F} \vee (Q \wedge (\sim P)) \rightarrow Q \Leftrightarrow (Q \wedge (\sim P)) \rightarrow Q \Leftrightarrow (\sim (Q \wedge (\sim P))) \vee Q$
$\Leftrightarrow (\sim Q \vee P) \vee Q \Leftrightarrow \mathbf{T} \vee (P \vee Q) \Leftrightarrow \mathbf{T}$.

Alternatively, we may consider the following truth table:

P	Q	$\sim P$	$P \vee Q$	$(P \vee Q) \wedge (\sim P)$	$((P \vee Q) \wedge (\sim P)) \rightarrow Q$
T	T	F	T	F	**T**
T	F	F	T	F	**T**
F	T	T	T	T	**T**
F	F	T	F	F	**T**

So the given statement is always true regardless of the truth values of P and Q.

(c) We only need to show that $((P \to Q) \wedge (Q \to R)) \to (P \to R) \Longleftrightarrow \mathbf{T}$:

$((P \to Q) \wedge (Q \to R)) \to (P \to R)$

$\Leftrightarrow \sim ((P \to Q) \wedge (Q \to R)) \vee (P \to R)$

$\Leftrightarrow (\sim (P \to Q) \vee \sim (Q \to R)) \vee (P \to R)$

$\Leftrightarrow ((P \wedge (\sim Q)) \vee (Q \wedge (\sim R))) \vee (P \to R)$

$\Leftrightarrow ((P \wedge (\sim Q)) \vee (Q \wedge (\sim R))) \vee (\sim P) \vee R$

$\Leftrightarrow ((P \wedge (\sim Q)) \vee (\sim P) \vee (Q \wedge (\sim R))) \vee R$

$\Leftrightarrow (\sim Q) \vee (\sim P) \vee Q \vee R$

$\Leftrightarrow \mathbf{T} \vee (\sim P) \vee R \Leftrightarrow \mathbf{T}$.

Alternatively, we may consider the following truth table:

P	Q	R	$P \to Q$	$Q \to R$	$(P \to Q) \wedge (Q \to R)$	$P \to R$	$(P \to Q) \wedge (Q \to R) \to (P \to R)$
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	F	T	F	T	**T**
T	F	F	F	T	F	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	F	F	T	**T**
F	F	T	T	T	T	T	**T**
F	F	F	T	T	T	T	**T**

So the given statement is always true regardless of the truth values of P, Q and R.

(d) (We normally do not use truth tables when there are 4 or more statements involved)

We only need to show that $((P \to Q) \wedge (R \to S) \wedge (P \vee R)) \to (Q \vee S) \Leftrightarrow \mathbf{T}$:

$((P \to Q) \wedge (R \to S) \wedge (P \vee R)) \to (Q \vee S)$

$\Leftrightarrow \sim [(P \to Q) \wedge (R \to S) \wedge (P \vee R)] \vee (Q \vee S)$

$\Leftrightarrow \sim [(\sim P \vee Q) \wedge (\sim R \vee S) \wedge (P \vee R)] \vee (Q \vee S)$

$\Leftrightarrow (P \wedge \sim Q) \vee (R \wedge \sim S) \vee (\sim P \wedge \sim R) \vee (Q \vee S)$

$\Leftrightarrow (P \wedge \sim Q) \vee Q \vee (R \wedge \sim S) \vee S \vee (\sim P \wedge \sim R)$

$\Leftrightarrow [(P \vee Q) \wedge (\sim Q \vee Q)] \vee [(R \vee S) \wedge (\sim S \vee S)] \vee (\sim P \wedge \sim R)$

$\Leftrightarrow [(P \vee Q) \wedge \mathbf{T}] \vee [(R \vee S) \wedge \mathbf{T}] \vee (\sim P \wedge \sim R)$

$\Leftrightarrow (P \vee Q) \vee (R \vee S) \vee (\sim P \wedge \sim R)$

$\Leftrightarrow (P \vee R) \vee (\sim P \wedge \sim R) \vee (Q \vee S)$

$\Leftrightarrow (P \vee R) \vee \sim (P \vee R) \vee (Q \vee S)$

$\Leftrightarrow \mathbf{T} \vee (Q \vee S)$

$\Leftrightarrow \mathbf{T}$

3. According to the given information, we have for any student x, $P(x) \Longrightarrow Q(x)$, and $\sim Q(x) \Longrightarrow R(x)$.

(a) Annie attends all the lectures, so P(Annie) is true. Then Q(Annie) is true, so she can finish at least 60% of the questions in the final exam correctly. However, no conclusion can be drawn on her grade.

(b) Since P(Bobby) is false, so we are not able to deduce the truth value of Q(Bobby) (and hence R(Bobby)).

(c) Given that Q(Calvin) is false, we can deduce that P(Calvin) is false, that is, Calvin attends (strictly) less than 80% of the lectures. Furthermore, we can deduce that R(Calvin) is true, that is, he gets at most a $C+$.

(d) No conclusion can be drawn on his attendance rate and his grade (since Q(David) is true).

(e) Given that R(Eddy) is false, we can deduce that $\sim Q$(Eddy) is false, in other words, Q(Eddy) is true. Hence Eddy finishes at least 60% of the questions in the final exam correctly. However, no conclusion can be drawn on his attendance rate.

(f) No conclusion can be drawn on her attendance rate and her number of correct answers in the final exam (since R(Fiona) is true).

4. (a) $\forall x, y \in \mathbb{Z},\ x + y \in \mathbb{Z}$.

   (b) $\forall x, y \in \mathbb{Q},\ xy \in \mathbb{Q}$.

   (c) $\forall a \in E,\ \exists x, y \in P,\ a = x + y$.

   (d) $\exists a \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ a + x = x$.

5. (a) "α is not an upper bound of A" means

$$\exists a \in A,\ a > \alpha.$$

   (b) "α is not a least upper bound of A" means

$$(\exists a \in A,\ a > \alpha)\ \underline{\text{or}}\ (\exists \text{ an upper bound } \alpha' \text{ of } A,\ \alpha > \alpha'.)$$

6. "A is not open" means

$$\exists x \in A,\ \forall \epsilon > 0,\ (x - \epsilon,\ x + \epsilon) \not\subseteq A,$$

   or more precisely,

$$\exists x \in A,\ \forall \epsilon > 0,\ \exists y \in (x - \epsilon,\ x + \epsilon) \text{ and } y \notin A.$$

7. (a) There exist $a, b \in \mathbb{R}$ such that $a \geq b$ and $a \neq b$ and $a \leq b$.

   (b) There exist $x, y \in (0, \infty)$ such that for any $n \in \mathbb{N}$, $nx \leq y$.

   (c) For any $N \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that $n \geq N$ and $\dfrac{1}{n^2} \geq \dfrac{2}{71}$.

8. (a) (i) There exists a vector $\underline{v}$ in $\mathbb{R}^3$ such that for any $c_1, c_2, c_3 \in \mathbb{R}$, $\underline{v} \neq c_1\underline{v}_1 + c_2\underline{v}_2 + c_3\underline{v}_3$.

   (ii) There exist $c_1, c_2, c_3 \in \mathbb{R}$ such that $c_1\underline{v}_1 + c_2\underline{v}_2 + c_3\underline{v}_3 = \underline{0}$ and ($c_1 \neq 0$ or $c_2 \neq 0$ or $c_3 \neq 0$). (A neater way of presenting (ii) is: There exist $c_1, c_2, c_3 \in \mathbb{R}$, **not all zero**, such that $c_1\underline{v}_1 + c_2\underline{v}_2 + c_3\underline{v}_3 = \underline{0}$.)

   (b) (i) There exists $x \in (a, b)$ and there exists $\epsilon > 0$ such that for any $\delta > 0$, there exists $y \in (a, b)$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon$.

   (ii) There exists $\epsilon > 0$ such that for any $\delta > 0$, there exist $x, y \in (a, b)$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon$.

9. We will show that

$$[\forall x \in S, (P(x) \longrightarrow Q(x))] \longrightarrow [(\forall x \in S, P(x)) \longrightarrow (\forall x \in S, Q(x))]$$

is logically equivalent to the tautology.

$$\begin{aligned}
& [\forall x \in S, (P(x) \longrightarrow Q(x))] \longrightarrow [(\forall x \in S, P(x)) \longrightarrow (\forall x \in S, Q(x))] \\
\iff & \sim [\forall x \in S, (P(x) \longrightarrow Q(x))] \vee [(\forall x \in S, P(x)) \longrightarrow (\forall x \in S, Q(x))] \\
\iff & [\exists x \in S, P(x) \wedge \sim Q(x)] \vee [(\exists x \in S, \sim P(x)) \vee (\forall x \in S, Q(x))] \\
\iff & (\exists x \in S, (P(x) \wedge \sim Q(x)) \vee \sim P(x)) \vee (\forall x \in S, Q(x)) \\
& \text{(here we used the result from } \textit{Tutorial 2, Q.5(a)}\text{)} \\
\iff & (\exists x \in S, \mathbf{T} \wedge (\sim Q(x) \vee \sim P(x))) \vee (\forall x \in S, Q(x)) \\
\iff & (\exists x \in S, (\sim Q(x) \vee \sim P(x))) \vee (\forall x \in S, Q(x)) \\
\iff & (\exists x \in S, \sim Q(x)) \vee (\exists x \in S, \sim P(x)) \vee (\forall x \in S, Q(x)) \\
& \text{(here we used the result from } \textit{Tutorial 2, Q.5(a)} \text{ again)} \\
\iff & (\exists x \in S, \sim P(x)) \vee (\exists x \in S, \sim Q(x)) \vee (\forall x \in S, Q(x)) \\
\iff & (\exists x \in S, \sim P(x)) \vee \mathbf{T} \\
\iff & \mathbf{T}
\end{aligned}$$

10. For convenience, we use A to represent the statement "$\forall x \in S, (P(x) \vee Q(x))$" and B to represent the statement "$\exists x \in S, \sim Q(x)$". Thus we are required to show that $A \vee B$ is a tautology.

    Consider the two cases (i) B is true and (ii) B is false.

    Case (i) B is true: Then $A \vee B$ must be true.

    Case (ii) B is false: This is logically equivalent to saying "$\forall x \in S, Q(x)$" is true. But this implies "$\forall x \in S, (P(x) \vee Q(x))$" is true (see also *Tutorial 2, Q.5(c)*). It follows that A is true and hence $A \vee B$ must be true.

    We can now conclude that $A \vee B$ is always true and hence it is a tautology.

11. (a) We first write the statement as

    For any $x, y \in \mathbb{Z}$, if x and y are even, then $x + y$ is even.

    Here is the proof:
    For any $x, y \in \mathbb{Z}$, if x and y are even, then we have $x = 2k_1$ and $y = 2k_2$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

    $$x + y = 2k_1 + 2k_2 = 2(k_1 + k_2).$$

    Since $k_1 + k_2 \in \mathbb{Z}$, so $x + y$ is also even.

    (b) We first write the statement as

    For any $x, y \in \mathbb{Z}$, if x and y are odd, then $x + y$ is even.

    Here is the proof:
    For any $x, y \in \mathbb{Z}$, if x and y are odd, then we have $x = 2k_1 + 1$ and $y = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

    $$x + y = (2k_1 + 1) + (2k_2 + 1) = 2(k_1 + k_2 + 1).$$

    Since $k_1 + k_2 + 1 \in \mathbb{Z}$, so $x + y$ is even.

(c) We first write the statement as

$$\text{For any } x, y \in \mathbb{Z}, \text{ if } x \text{ is even and } y \text{ is odd, then } x + y \text{ is odd.}$$

Here is the proof:
For any $x, y \in \mathbb{Z}$, if x is even and y is odd, then we have $x = 2k_1$ and $y = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

$$x + y = 2k_1 + (2k_2 + 1) = 2(k_1 + k_2) + 1.$$

Since $k_1 + k_2 \in \mathbb{Z}$, so $x + y$ is odd.

(d) We first write the statement as

$$\text{For any } x, y \in \mathbb{Z}, \text{ if } x \text{ and } y \text{ are even, then } xy \text{ is even.}$$

Here is the proof:
For any $x, y \in \mathbb{Z}$, if x and y are even, then we have $x = 2k_1$ and $y = 2k_2$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

$$xy = (2k_1)(2k_2) = 2(2k_1k_2).$$

Since $2k_1k_2 \in \mathbb{Z}$, so xy is also even.

(e) We first write the statement as

$$\text{For any } x, y \in \mathbb{Z}, \text{ if } x \text{ and } y \text{ are odd, then } xy \text{ is odd.}$$

Here is the proof:
For any $x, y \in \mathbb{Z}$, if x and y are odd, then we have $x = 2k_1 + 1$ and $y = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

$$xy = (2k_1 + 1)(2k_2 + 1) = 2(2k_1k_2 + k_1 + k_2) + 1.$$

Since $2k_1k_2 + k_1 + k_2 \in \mathbb{Z}$, so xy is odd.

(f) We first write the statement as

$$\text{For any } x, y \in \mathbb{Z}, \text{ if } x \text{ is even and } y \text{ is odd, then } xy \text{ is even.}$$

Here is the proof:
For any $x, y \in \mathbb{Z}$, if x is even and y is odd, then we have $x = 2k_1$ and $y = 2k_2 + 1$ for some $k_1, k_2 \in \mathbb{Z}$. It follows that

$$xy = 2k_1(2k_2 + 1).$$

Since $k_1(2k_2 + 1) \in \mathbb{Z}$, so xy is even.

12. (a) We first write the statement as

$$\text{For any } x, y \in \mathbb{R}, \text{ if } x, y \in \mathbb{Q}, \text{ then } x + y \in \mathbb{Q}.$$

Here is the proof:
For any $x, y \in \mathbb{R}$, if $x, y \in \mathbb{Q}$, then we have $x = \frac{p_1}{q_1}$ and $y = \frac{p_2}{q_2}$ for some $p_1, q_1, p_2, q_2 \in \mathbb{Z}$ and $q_1, q_2 \neq 0$. It follows that

$$x + y = \frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2}.$$

Since $p_1 q_2 + p_2 q_1, q_1 q_2 \in \mathbb{Z}$ and $q_1 q_2 \neq 0$, so $x + y \in \mathbb{Q}$.

(b) We first write the statement as

$$\text{For any } x, y \in \mathbb{R}, \text{ if } x, y \in \mathbb{Q}, \text{ then } xy \in \mathbb{Q}.$$

Here is the proof:
For any $x, y \in \mathbb{R}$, if $x, y \in \mathbb{Q}$, then we have $x = \frac{p_1}{q_1}$ and $y = \frac{p_2}{q_2}$ for some $p_1, q_1, p_2, q_2 \in \mathbb{Z}$ and $q_1, q_2 \neq 0$. It follows that

$$xy = \frac{p_1}{q_1} \cdot \frac{p_2}{q_2} = \frac{p_1 p_2}{q_1 q_2}.$$

Since $p_1 p_2, q_1 q_2 \in \mathbb{Z}$ and $q_1 q_2 \neq 0$, so $xy \in \mathbb{Q}$.

(c) We first write the statement as

$$\text{For any } x, y \in \mathbb{R}, \text{ if } x, y \in \mathbb{Q} \text{ and } y \neq 0, \text{ then } \frac{x}{y} \in \mathbb{Q}.$$

Here is the proof:
For any $x, y \in \mathbb{R}$, if $x, y \in \mathbb{Q}$ and $y \neq 0$, then we have $x = \frac{p_1}{q_1}$ and $y = \frac{p_2}{q_2}$ for some $p_1, q_1, p_2, q_2 \in \mathbb{Z}$ and $p_2, q_1, q_2 \neq 0$. It follows that

$$\frac{x}{y} = \frac{\frac{p_1}{q_1}}{\frac{p_2}{q_2}} = \frac{p_1 q_2}{q_1 p_2}.$$

Since $p_1 q_2, q_1 p_2 \in \mathbb{Z}$ and $q_1 p_2 \neq 0$, so $\frac{x}{y} \in \mathbb{Q}$.

13. (a) For any $a, b, c \in \mathbb{Z}$, if $a|b$ and $b|c$, then there exists $k_1 \in \mathbb{Z}$ such that $b = ak_1$, and there exists $k_2 \in \mathbb{Z}$ such that $c = bk_2$. So we have $c = a(k_1 k_2)$ for some $k_1 k_2 \in \mathbb{Z}$ and hence $a|c$.

(b) **"$\Longrightarrow$" ("only if" part)**: For any $a, b \in \mathbb{Z} \setminus \{0\}$, assume that $a|b$ and $b|a$. By definition, there exists $c \in \mathbb{Z}$ such that $b = ac$, and there exists $d \in \mathbb{Z}$ such that $a = db$. Plugging this last equation into the previous one yields $b = (db)c$. Since $b \neq 0$, it follows that $dc = 1$. Note that c and d are integers, the equation $dc = 1$ implies that $c = d = 1$ or $c = d = -1$. In the former case, we have $a = b$ and in the latter case $a = -b$.

"$\Longleftarrow$" ("if" part): For any $a, b \in \mathbb{Z} \setminus \{0\}$, asssume that $a = b$ or $a = -b$. If $a = b$, then we take $c = 1 \in \mathbb{Z}$ so that $b = ac$ and hence we have $a|b$. Moreover, we can take $d = 1 \in \mathbb{Z}$ so that $a = bd$ and hence we have $b|a$. Similar argument applies to the case $a = -b$ where we can take $c = -1$ and $d = -1$ so that $a|b$ and $b|a$.

14. (a) First of all, for any $x, y \in \mathbb{R}$, if x and y are positive, then $\sqrt{x}$, $\sqrt{y}$ and $\sqrt{xy}$ are all well-defined. Now observe that for any $x, y \in \mathbb{R}$, if x and y are positive, we must have $(\sqrt{x} - \sqrt{y})^2 \geq 0$. It then follows that

$$(\sqrt{x} - \sqrt{y})^2 \geq 0 \implies x - 2\sqrt{xy} + y \geq 0 \implies x + y \geq 2\sqrt{xy} \implies \sqrt{xy} \leq \frac{x+y}{2}.$$

(b) For any positive real numbers x and y,

$$\begin{aligned} \sqrt{xy} = \frac{x+y}{2} &\iff x - 2\sqrt{xy} + y = 0 \\ &\iff (\sqrt{x} - \sqrt{y})^2 = 0 \\ &\iff \sqrt{x} = \sqrt{y} \\ &\iff x = y. \end{aligned}$$

15. By Question **14(a)**, we have

$$a + b \geq 2\sqrt{ab}, \tag{1}$$
$$b + c \geq 2\sqrt{bc}, \tag{2}$$
$$a + c \geq 2\sqrt{ac}. \tag{3}$$

From the above inequalities, we get $(a+b)(b+c)(a+c) \geq (2\sqrt{ab})(2\sqrt{bc})(2\sqrt{ac}) = 8abc$. The equality holds if and only if equalities in (1), (2) and (3) hold. Then by Question **14(b)**, equalities in (1), (2) and (3) hold if and only if $a = b = c$. Hence the result.

Alternative Solution:

Note that for any $a, b, c \in \mathbb{R}$, if a, b, c are positive, then

$$\begin{aligned} & (a+b)(b+c)(c+a) - 8abc \\ = \; & (ab + ac + b^2 + bc)(c+a) - 8abc \\ = \; & (abc + ac^2 + b^2c + bc^2 + a^2b + a^2c + ab^2 + abc) - 8abc \\ = \; & a(b^2 - 2bc + c^2) + b(a^2 - 2ac + c^2) + c(a^2 - 2ab + b^2) \\ = \; & a(b-c)^2 + b(a-c)^2 + c(a-b)^2 \\ \geq \; & 0 \quad (\text{as } a, b, c > 0 \text{ and } (b-c)^2, (a-c)^2, (a-b)^2 \geq 0), \end{aligned}$$

and the equality holds iff $(b-c)^2 = (a-c)^2 = (a-b)^2 = 0$ iff $a = b = c$.

16. First of all, note that for any $a, b, c, x \in \mathbb{R}$ with $a \neq 0$, we have

$$\begin{aligned} ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) \\ &= a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a}\right] \\ &= a\left[\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2}\right]. \end{aligned} \tag{4}$$

Thus,

$$\begin{aligned} ax^2+bx+c=0 &\overset{(4)}{\iff} a\left[\left(x+\frac{b}{2a}\right)^2-\frac{b^2-4ac}{4a^2}\right]=0 \\ &\iff \left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2} \\ &\iff x+\frac{b}{2a}=\pm\frac{\sqrt{b^2-4ac}}{2a} \\ &\iff x=-\frac{b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}. \end{aligned}$$

17. For any $a, b, c \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, suppose we have $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, that is, $a-b = nk_1$ and $b-c = nk_2$ for some integers k_1, k_2. Then $a-c = (a-b)+(b-c) = n(k_1+k_2)$ and hence $a \equiv c \pmod{n}$.

18. For any $a, b, c \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, suppose we have $a \equiv b \pmod{n}$, that is, $a - b = nk$ for some integer k. Then $ca - cb = cnk = n(ck)$ and hence $ca \equiv cb \pmod{n}$.

19. For any $a, b, c \in \mathbb{Z}$ and for any $n \in \mathbb{N}$ with $n \geq 2$, suppose we have $a \equiv b \pmod{n}$, that is, $a - b = nk$ for some integer k. Then

$$a^2 - b^2 = (a-b)(a+b) = kn(a+b) = n(ka+kb)$$

and hence $a^2 \equiv b^2 \pmod{n}$.

20. For any $a, b \in \mathbb{Z}$, if $a \equiv b \pmod{6}$, then we have $a - b = 6k$ for some integer k. But we can also write $a - b = 6k$ as $a - b = 3(2k)$, which implies $a \equiv b \pmod{3}$.

21. (a) For any x,

$$\begin{aligned} x \in (A \cup B) \setminus C &\iff x \in (A \cup B) \text{ and } x \notin C \\ &\iff ((x \in A) \text{ or } (x \in B)) \text{ and } (x \notin C) \\ &\iff ((x \in A) \text{ and } (x \notin C)) \text{ or } ((x \in B) \text{ and } (x \notin C)) \\ &\iff (x \in A \setminus C) \text{ or } (x \in B \setminus C) \\ &\iff x \in (A \setminus C) \cup (B \setminus C). \end{aligned}$$

So $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

(b) For any x,

$$\begin{aligned} x \in A \setminus (B \cup C) &\iff (x \in A) \text{ and } (x \notin (B \cup C)) \\ &\iff (x \in A) \text{ and } ((x \notin B) \text{ and } (x \notin C)) \quad \text{(De Morgan's law)} \\ &\iff (x \in A) \text{ and } (x \notin B) \text{ and } (x \notin C) \\ &\iff ((x \in A) \text{ and } (x \notin B)) \text{ and } ((x \in A) \text{ and } (x \notin C)) \\ &\iff (x \in A \setminus B) \text{ and } (x \in A \setminus C) \\ &\iff x \in (A \setminus B) \cap (A \setminus C). \end{aligned}$$

So $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

(c) For any x, y,

$$\begin{aligned}
(x,y) \in (A \times B) \cap (C \times D) &\iff (x,y) \in A \times B \text{ and } (x,y) \in C \times D \\
&\iff (x \in A \text{ and } y \in B) \text{ and } (x \in C \text{ and } y \in D) \\
&\iff (x \in A \text{ and } x \in C) \text{ and } (y \in B \text{ and } y \in D) \\
&\iff x \in (A \cap C) \text{ and } y \in (B \cap D) \\
&\iff (x,y) \in (A \cap C) \times (B \cap D).
\end{aligned}$$

So $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

22. (a) For any subsets A, B, C of a universal set U,

$$A \setminus B = \{x \in U : x \in A \text{ and } x \notin B\} = \{x \in U : x \in A \text{ and } x \in B^c\} = A \cap B^c.$$

(b) First of all, we note that

$$B \triangle C = (B \setminus C) \cup (C \setminus B) \overset{\text{by (a)}}{=} (B \cap C^c) \cup (C \cap B^c) = (B \cap C^c) \cup (B^c \cap C).$$

Therefore,

$$\begin{aligned}
A \cap (B \triangle C) &= A \cap ((B \cap C^c) \cup (B^c \cap C)) \\
&= (A \cap B \cap C^c) \cup (A \cap B^c \cap C).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
& (A \cap B) \triangle (A \cap C) \\
= \; & [(A \cap B) \cap (A \cap C)^c] \cup [(A \cap B)^c \cap (A \cap C)] \\
= \; & [(A \cap B) \cap (A^c \cup C^c)] \cup [(A^c \cup B^c) \cap (A \cap C)] \\
= \; & [(A \cap B \cap A^c) \cup (A \cap B \cap C^c)] \cup [(A^c \cap A \cap C) \cup (B^c \cap A \cap C)] \\
= \; & [\emptyset \cup (A \cap B \cap C^c)] \cup [\emptyset \cup (B^c \cap A \cap C)] \\
= \; & (A \cap B \cap C^c) \cup (A \cap B^c \cap C).
\end{aligned}$$

The result follows.

23. Statement (i) is **false** and statement (ii) is **true**.

<u>Proof for (ii)</u>: For any sets A, B, C, D, assume that $A \subseteq C$ and $B \subseteq D$. We want to show that $A \cap B \subseteq C \cap D$. For any $x \in A \cap B$, we have

$$\begin{aligned}
x \in A \cap B &\implies x \in A \text{ and } x \in B \\
&\implies x \in C \text{ and } x \in D \text{ (by assumption)} \\
&\implies x \in C \cap D.
\end{aligned}$$

<u>Examples for (i)</u>: To show that (i) is false, we need to find example of sets A, B, C, D so that $A \cap B \subseteq C \cap D$ and ($A \not\subseteq C$ or $B \not\subseteq D$). Note that it is not necessary to give an example with both $A \not\subseteq C$ and $B \not\subseteq D$

Let $A = \{1\}$, $B = \{2\}$, $C = \{2\}$ and $D = \{2\}$. Then $A \cap B = \emptyset$, which is certainly a subset of $C \cap D = \{2\}$. However, A is not a subset of C as $1 \in A$ and $1 \notin C$.

24. (a) **"$\Longrightarrow$" ("only if" part)**: Suppose $A \subseteq B$. For any X,

$$\begin{aligned} X \in \mathcal{P}(A) &\Longrightarrow X \subseteq A \\ &\Longrightarrow X \subseteq B \quad \text{(since } A \subseteq B \text{ by assumption)} \\ &\Longrightarrow X \in \mathcal{P}(B) \end{aligned}$$

Hence we have $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

"$\Longleftarrow$" ("if" part): Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

<u>*Solution 1*</u>

Note that for any set X, we have $X \in \mathcal{P}(X)$. Thus,

$$\begin{aligned} A \in \mathcal{P}(A) &\Longrightarrow A \in \mathcal{P}(B) \quad \text{(since } \mathcal{P}(A) \subseteq \mathcal{P}(B) \text{ by assumption)} \\ &\Longrightarrow A \subseteq B. \end{aligned}$$

<u>*Solution 2*</u>

For any x,

$$\begin{aligned} x \in A &\Longrightarrow \{x\} \subseteq A \\ &\Longrightarrow \{x\} \in \mathcal{P}(A) \\ &\Longrightarrow \{x\} \in \mathcal{P}(B) \quad \text{(since } \mathcal{P}(A) \subseteq \mathcal{P}(B) \text{ by assumption)} \\ &\Longrightarrow \{x\} \subseteq B \\ &\Longrightarrow x \in B \end{aligned}$$

Hence we have $A \subseteq B$.

(b) For any X,

$$\begin{aligned} X \in \mathcal{P}(A) \cap \mathcal{P}(B) &\Longleftrightarrow X \in \mathcal{P}(A) \text{ and } X \in \mathcal{P}(B) \\ &\Longleftrightarrow X \subseteq A \text{ and } X \subseteq B \\ &\Longleftrightarrow X \subseteq A \cap B \\ &\Longleftrightarrow X \in \mathcal{P}(A \cap B). \end{aligned}$$

Hence we have $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.

25. To show $A = B$, we divide the proof into two parts: (i) $A \subseteq B$ and (ii) $B \subseteq A$:

(i) Suppose we have $(x, y) \in A$. Then we get $x \equiv y \pmod 6$, that is, $x - y = 6k$ for some $k \in \mathbb{Z}$. As $x - y = 6k = 2(3k) = 3(2k)$, this implies $x \equiv y \pmod 2$ and $x \equiv y \pmod 3$. It follows that $(x, y) \in B$.

(ii) Suppose we have $(x, y) \in B$. To show $(x, y) \in A$, it suffices to show $x - y = 6k'$ for some $k' \in \mathbb{Z}$. As $(x, y) \in B$, we get $x \equiv y \pmod 2$ and $x \equiv y \pmod 3$, that is,

$$x - y = 2k_1 \text{ and } x - y = 3k_2 \text{ for some } k_1, k_2 \in \mathbb{Z}.$$

This implies $2k_1 = 3k_2$, and so $3k_2$ is even. It follows that k_2 must be even as if k_2 is odd, then $3k_2$ is also odd, which contradicts the fact that $3k_2$ is even. Hence we can write $k_2 = 2k'$ for some $k' \in \mathbb{Z}$ and so we get $x - y = 3k_2 = 6k'$, which is what we want to show.

26. We prove the statement by contrapositive.

Assume that $n \not\equiv 1 \pmod 6$ and $n \not\equiv 5 \pmod 6$. Now we need to show that n is not prime. By assumption, we have $n \equiv 0 \pmod 6$ or $n \equiv 2 \pmod 6$ or $n \equiv 3 \pmod 6$ or $n \equiv 4 \pmod 6$.

<u>Case 1</u>: If $n \equiv 0 \pmod 6$, then $n = 6k$ for some $k \in \mathbb{N}$ (k cannot be non-positive because $n > 3$). So n is not prime.

<u>Case 2</u>: If $n \equiv 2 \pmod 6$, then $n = 6k + 2 = 2(3k + 1)$ for some $k \in \mathbb{N}$ (k cannot be non-positive because $n > 3$). So n is not prime.

<u>Case 3</u>: If $n \equiv 3 \pmod 6$, then $n = 6k + 3 = 3(2k + 1)$ for some $k \in \mathbb{N}$ (k cannot be non-positive because $n > 3$). So n is not prime.

<u>Case 4</u>: If $n \equiv 4 \pmod 6$, then $n = 6k + 4 = 2(3k + 2)$ for some $k \in \mathbb{N}$ (k cannot be non-positive because $n > 3$). So n is not prime.

For all cases, n is not prime and therefore the result follows.

27. **"$\Longrightarrow$" ("only if" part)**: We prove the "only if" part by contrapositve.

Assume that $n \equiv 0 \pmod 3$ **or** $n \equiv 1 \pmod 3$. Now we need to show that $n^2 \equiv n \pmod 3$.

<u>Case 1</u>: If $n \equiv 0 \pmod 3$, then $n = 3k$ for some $k \in \mathbb{Z}$ and hence

$$n^2 - n = n(n-1) = 3k(3k-1).$$

So $n^2 \equiv n \pmod 3$. Alternatively, we may apply the result in **Q.19** to get

$$n \equiv 0 \pmod 3 \implies n^2 \equiv 0^2 = 0 \pmod 3$$

and hence $n^2 \equiv n \ (\equiv 0) \pmod 3$.

<u>Case 2</u>: If $n \equiv 1 \pmod 3$, then $n = 3k + 1$ for some $k \in \mathbb{Z}$ and hence

$$n^2 - n = n(n-1) = 3k(3k+1).$$

So $n^2 \equiv n \pmod 3$. Alternatively, we may apply the result in **Q.19** to get

$$n \equiv 1 \pmod 3 \implies n^2 \equiv 1^2 = 1 \pmod 3$$

and hence $n^2 \equiv n \ (\equiv 1) \pmod 3$.

For all cases, $n^2 \equiv n \pmod 3$ and therefore the "only if" part follows.

"$\Longleftarrow$" ("if" part): Assume that $n \not\equiv 0 \pmod 3$ and $n \not\equiv 1 \pmod 3$. Then it must be the case that $n \equiv 2 \pmod 3$, and hence $n = 3k + 2$ for some $k \in \mathbb{Z}$. Then

$$n^2 - n = n(n-1) = (3k+2)(3k+1) = 9k^2 + 9k + 2 = 3(3k^2 + 3k) + 2$$

and hence $n^2 \equiv n + 2 \not\equiv n \pmod 3$.

28. We will prove that "n is even" is a necessary and sufficient condition for $5^n \equiv 1 \pmod 8$, in other words, we will prove that for any nonnegative integer n,

$$5^n \equiv 1 \pmod 8 \iff n \text{ is even.}$$

"$\Longrightarrow$" ("only if" part): We prove the "only if" part by contrapositve.

Assume that n is odd. Then $n = 2k + 1$ for some nonnegative integer k and hence

$$5^n = 5^{2k+1} = 25^k \cdot 5 \equiv 1 \cdot 5 = 5 \pmod 8.$$

Here we have used the fact that $25 \equiv 1 \pmod 8$ and hence any (nonnegative) powers of 25 will also be congruent to 1 (mod 8). Thus we have proved that if n is odd, then $5^n \not\equiv 1 \pmod 8$.

"$\Longleftarrow$" ("if" part): Assume that n is even. Then $n = 2k$ for some nonnegative integer k and hence $5^n = 5^{2k} = 25^k \equiv 1 \pmod 8$.

Remark: In general, if you do not have any idea about the problem initially, you may try to plug in some small values and compute a few examples. Examples can often guide you to see a pattern. In this question, if you try to compute the first few values for 5^n, you will get $5^0 = 1$, $5^1 = 5$, $5^2 = 25 \equiv 1 \pmod 8$, $5^3 = 125 \equiv 5 \pmod 8$, $5^4 = 625 \equiv 1 \pmod 8$. Thus from the first few cases, we already observe that the odd powers of 5 seem to be always congruent to 5 (mod 8) and the even powers of 5 seem to be always congruent to 1 (mod 8).

29. (a) For any $n \in \mathbb{Z}$, we have

<u>Case 1</u>: $n \equiv 0 \pmod 3$. Then $n = 3k$ for some $k \in \mathbb{Z}$ and so

$$n^2 = 9k^2 = 3(3k^2) \equiv 0 \pmod 3.$$

Alternatively, we may apply the result in **Q.19** to get

$$n \equiv 0 \pmod 3 \implies n^2 \equiv 0^2 = 0 \pmod 3.$$

<u>Case 2</u>: $n \equiv 1 \pmod 3$. Then $n = 3k + 1$ for some $k \in \mathbb{Z}$ and so

$$n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1 \equiv 1 \pmod 3.$$

Alternatively, we may apply the result in **Q.19** to get

$$n \equiv 1 \pmod 3 \implies n^2 \equiv 1^2 = 1 \pmod 3.$$

<u>Case 3</u>: $n \equiv 2 \pmod 3$. Then $n = 3k + 2$ for some $k \in \mathbb{Z}$ and so

$$n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1 \equiv 0 \pmod 3.$$

Alternatively, we may apply the result in **Q.19** to get

$$n \equiv 2 \pmod 3 \implies n^2 \equiv 2^2 = 4 \equiv 1 \pmod 3.$$

(b) We shall apply the result in **Q.19**:
For any $n \in \mathbb{Z}$, we have
<u>Case 1</u>: $n \equiv 0 \pmod{4} \implies n^2 \equiv 0^2 = 0 \pmod{4}$.
<u>Case 2</u>: $n \equiv 1 \pmod{4} \implies n^2 \equiv 1^2 = 1 \pmod{4}$.
<u>Case 3</u>: $n \equiv 2 \pmod{4} \implies n^2 \equiv 2^2 = 4 \equiv 0 \pmod{4}$.
<u>Case 4</u>: $n \equiv 3 \pmod{4} \implies n^2 \equiv 3^2 = 9 \equiv 1 \pmod{4}$.
Thus for any $n \in \mathbb{Z}$, $n^2 \equiv 0$ or $1 \pmod 4$.

(c) We shall apply the result in **Q.19**:
For any $n \in \mathbb{Z}$, we have
<u>Case 1</u>: $n \equiv 0 \pmod{5} \implies n^2 \equiv 0^2 = 0 \pmod{5}$.
<u>Case 2</u>: $n \equiv 1 \pmod{5} \implies n^2 \equiv 1^2 = 1 \pmod{5}$.
<u>Case 3</u>: $n \equiv 2 \pmod{5} \implies n^2 \equiv 2^2 = 4 \pmod{5}$.
<u>Case 4</u>: $n \equiv 3 \pmod{5} \implies n^2 \equiv 3^2 = 9 \equiv 4 \pmod{5}$.
<u>Case 5</u>: $n \equiv 4 \pmod{5} \implies n^2 \equiv 4^2 = 16 \equiv 1 \pmod{5}$.
Thus for any $n \in \mathbb{Z}$, $n^2 \equiv 0$ or 1 or $4 \pmod 5$.

(d) We will show that for any $n \in \mathbb{Z}$, (i) $n^2 \equiv 0$ or 1 or 3 or $4 \pmod 6$; and (ii) $n^2 \equiv 0$ or 1 or 2 or $4 \pmod 7$. We shall apply the result in **Q.19**:

(i) For any $n \in \mathbb{Z}$, we have
<u>Case 1</u>: $n \equiv 0 \pmod{6} \implies n^2 \equiv 0^2 = 0 \pmod{6}$.
<u>Case 2</u>: $n \equiv 1 \pmod{6} \implies n^2 \equiv 1^2 = 1 \pmod{6}$.
<u>Case 3</u>: $n \equiv 2 \pmod{6} \implies n^2 \equiv 2^2 = 4 \pmod{6}$.
<u>Case 4</u>: $n \equiv 3 \pmod{6} \implies n^2 \equiv 3^2 = 9 \equiv 3 \pmod{6}$.
<u>Case 5</u>: $n \equiv 4 \pmod{6} \implies n^2 \equiv 4^2 = 16 \equiv 4 \pmod{6}$.
<u>Case 6</u>: $n \equiv 5 \pmod{6} \implies n^2 \equiv 5^2 = 25 \equiv 1 \pmod{6}$.
Thus for any $n \in \mathbb{Z}$, $n^2 \equiv 0$ or 1 or 3 or $4 \pmod 6$.

(ii) For any $n \in \mathbb{Z}$, we have
<u>Case 1</u>: $n \equiv 0 \pmod{7} \implies n^2 \equiv 0^2 = 0 \pmod{7}$.
<u>Case 2</u>: $n \equiv 1 \pmod{7} \implies n^2 \equiv 1^2 = 1 \pmod{7}$.
<u>Case 3</u>: $n \equiv 2 \pmod{7} \implies n^2 \equiv 2^2 = 4 \pmod{7}$.
<u>Case 4</u>: $n \equiv 3 \pmod{7} \implies n^2 \equiv 3^2 = 9 \equiv 2 \pmod{7}$.
<u>Case 5</u>: $n \equiv 4 \pmod{7} \implies n^2 \equiv 4^2 = 16 \equiv 2 \pmod{7}$.
<u>Case 6</u>: $n \equiv 5 \pmod{7} \implies n^2 \equiv 5^2 = 25 \equiv 4 \pmod{7}$.
<u>Case 7</u>: $n \equiv 6 \pmod{7} \implies n^2 \equiv 6^2 = 36 \equiv 1 \pmod{7}$.
Thus for any $n \in \mathbb{Z}$, $n^2 \equiv 0$ or 1 or 2 or $4 \pmod 7$.

30. From the result of **Q.29(b)**, for any $m, n \in \mathbb{Z}$, m^2, n^2 can only be congruent to 0 or 1 (mod 4). Therefore, $m^2 - n^2$ can only be congruent to

$$0 - 0 = 0 \quad \text{or} \quad 1 - 0 = 1 \quad \text{or} \quad 0 - 1 = -1 \equiv 3 \quad \text{or} \quad 1 - 1 = 0 \pmod 4,$$

that is, $m^2 - n^2 \equiv 0$ or 1 or $3 \pmod 4$. However, $2022 \equiv 2 \pmod 4$. It follows that $m^2 - n^2$ can never be equal to 2022 for any $m, n \in \mathbb{Z}$.

31. One way to prove it is by showing that $n^7 \equiv n \pmod 7$ for each case $n \equiv 0, 1, 2, 3, 4, 5, 6 \pmod 7$. Here we will adopt another approach. Note that

$$n^7 - n \;=\; n(n^6 - 1) \;=\; n(n^3 - 1)(n^3 + 1) \;=\; n(n-1)(n+1)(n^2 + n + 1)(n^2 - n + 1).$$

It is easily seen that for any $n \in \mathbb{Z}$, if $n \equiv 0 \pmod 7$ or $n \equiv 1 \pmod 7$ or $n \equiv 6 \pmod 7$, then the right hand side of the above equation will be congruent to $0 \pmod 7$, and so $n^7 - n \equiv 0 \pmod 7$ when $n \equiv 0$ or 1 or $6 \pmod 7$. It remains to check the other four cases:

- If $n \equiv 2 \pmod 7$, then $n^2 + n + 1 \equiv 2^2 + 2 + 1 \equiv 7 \equiv 0 \pmod 7$.
- If $n \equiv 3 \pmod 7$, then $n^2 - n + 1 \equiv 3^2 - 3 + 1 \equiv 7 \equiv 0 \pmod 7$.
- If $n \equiv 4 \pmod 7$, then $n^2 + n + 1 \equiv 4^2 + 4 + 1 \equiv 21 \equiv 0 \pmod 7$.
- If $n \equiv 5 \pmod 7$, then $n^2 - n + 1 \equiv 5^2 - 5 + 1 \equiv 21 \equiv 0 \pmod 7$.

Hence in all cases we have $n^7 - n = n(n-1)(n+1)(n^2 - n + 1)(n^2 + n + 1) \equiv 0 \pmod 7$. The result follows.